# Yuvraj Sandhu

U.S. Citizen
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## Education

**McMaster University** Hamilton, ON
*Computer Engineering & Business Management* *Expected December 2026*
**Relevant Coursework**: High-Performance Programming, Introduction to Control Systems, Algorithm Design and Analysis, Software Development, Digital Systems Design, and Embedded Systems.

## Skills Summary

- **Languages**: C, C++, C#, Python, Java, SQL, VHDL, and JavaScript.
- **Embedded & Firmware**: PIC24F, STM32, nRF52 (nRF Connect SDK), ESP32, Zephyr RTOS, ROS, DFU and remote flashing, ICSP/J-Link, SPI, I2C, UART, and ONFI parallel NAND.
- **Wireless & Networking**: BLE 5.0 and BLE Mesh, LTE, WiFi, NFC, TCP/UDP, DHCP, and Software-Defined Radio (SDR).
- **Software & DevOps**: Linux, Windows, .NET Framework, multithreading, Qt, Git, CI/CD (Azure Pipelines), Pandas, and NumPy.
- **Test & Validation**: Hardware-in-the-loop (HIL) testing, LabVIEW, NI TestStand, Signal Hound spectrum analyzers, Keysight DAQ970A, oscilloscopes, RF power and frequency testing, failure analysis, and PCB rework.
- **Hardware & PCB Design**: Altium Designer (schematics, multilayer layout, BOM), wiring harnesses and interconnects, manufacturing and assembly drawings, KiCad, OrCAD PSpice, LTspice, Autodesk Fusion, and AutoCAD.

## Experience

**Recon Intelligence Corporation** Calgary, AB

*Embedded Hardware Engineer Intern* *June 2026 – Present*

- Designed and developed a complete **USB-powered NAND flash data recovery device** around a PIC24FJ microcontroller, owning schematic capture, design verification, BOM selection, and **four-layer PCB layout** in Altium Designer, prioritizing high-speed communication.
- Wrote **embedded C firmware** for the device's ONFI parallel NAND interface, optimizing timing-critical bit-banged read sequences to maximize data throughput.
- Verified clocking, power, and USB device configuration against manufacturer datasheets, and dispositioned a fifteen-item hardware **design review** with written justifications and corrective changes across power, protection, and signal integrity.
- Designed **wiring harnesses** and **electrical schematics** for mining machinery, along with wiring interconnects for strain gauges.
- Created **manufacturing and assembly drawings** to guide production of company hardware.

**JDRF Electromag Engineering Inc.** Mississauga, ON

*System Validation Engineer (Part-Time Contract)* *Jan 2026 – Feb 2026*

- Integrated CI/CD pipelines with J-Link flash testing to automate **firmware deployment and validation**, boosting test bench throughput and supporting on-time order fulfillment.
- Built remote-flashing and **device firmware update (DFU)** libraries for STM32 and nRF microcontrollers.
- Refined Azure-based firmware pipelines and workflows to automatically locate and identify failing deployed products.
- Traced prototype **defects to root cause** by analyzing large-scale test data against hardware schematics, defining error tolerances for end-to-end platform assurance.
- Performed **PCB analysis** and rework soldering for RMA repairs and defect isolation.

**JDRF Electromag Engineering Inc.** Mississauga, ON

*Research & Design Lab Engineering Intern* *June 2025 – Jan 2026*

- Designed and deployed a multithreaded **C# (.NET Framework)** Windows GUI application to test WiFi/LTE routers over DHCP, TCP, and UDP, making end-to-end validation **10× faster**.
- Wrote **Python** test scripts for multiple router verification tests running on **Linux**, and created **Azure Pipelines** to automatically and remotely flash routers during testing.
- Integrated oscilloscopes, a Signal Hound SA44B spectrum analyzer, a Keysight DAQ970A, and AC power supplies with the application over TCP using Python, LabVIEW, and NI TestStand, enabling **hardware-in-the-loop (HIL)** QA testing, NFC and RF product characterization, and automated product data analysis.
- Wrote RF test firmware and libraries in Python and C++ that set the LTE modem's and BLE radio's transmit power (dBm) and frequency across all 2.4 GHz BLE channels and the 1.9 GHz cellular band for automated **RF testing**.
- Co-developed the router test plan and helped design its **RF test fixture**, running programmatic RF tests on equipment to select fixture components.

- Designed and deployed a second multithreaded **C#** application for HIL testing of Bluetooth sensor modules over **NFC and UART**, built for real-time systems requiring precise timing.
- Developed **I2C and SPI** communications for the sensor module PCB, and designed a data communication structure that packs **image data over NFC** for RMSE analysis and validation.
- Set automated pass/fail limits through **statistical analysis** of test data in Python and Excel, and wrote **SQL** queries to track and update a database of tested products.
- Authored reusable **Python and LabVIEW API libraries**, version-controlled in Git, for the customer-facing Field Application Engineering team, and contributed to public Python libraries to improve operational efficiency.

**Toronto Hydro** Toronto, ON

*Data Analytics Intern* *May 2023 – Aug 2024*

- Developed a **Python** automation application with a **Qt** GUI that automatically gathers and sources project data through **SQL** queries and processes it in Pandas and NumPy pipelines, reducing manual data entry by 50%.
- Implemented **statistical models** and trend analysis in Python and Excel to forecast Light Rail Transit (LRT) project costs, developing predictive algorithms that improved planning accuracy and financial oversight.
- Performed load calculations, design reviews, and cost estimations for multi-million dollar infrastructure projects.

**The Canadian Remembrance Torch** Hamilton, ON

*Research & Design Lab Engineering Intern* *Oct 2021 – Nov 2022*

- Designed an **Arduino Nano**-based **embedded system** end to end, from custom PCB to **C++ firmware**, for **real-time** heart rate acquisition and LED-array visualization, with an on-device ML model classifying signal quality to reject noisy readings.
- Implemented power management algorithms and refined sensor calibration to achieve a 60% increase in battery life and ensure reliable operation under thermal stress.
- Designed the torch's outer body and shell in **Autodesk Fusion** and **AutoCAD** for outdoor and ergonomic use.

**MAV Woodworking** Bolton, ON

*Software Engineer* *June 2019 – Sept 2020*

- Developed **post-processors in Java** for SCM CNC machines, translating CAM toolpaths into machine-specific code.
- Wrote machining programs and macros in SCM's C#-based **Maestro Scripting Language (MSL)**, defining workpieces, geometry, and drilling and routing operations in code.
- Created **Python** interfaces to SCM Maestro CAD/CAM software to support custom designs.
- Designed custom parts and ran machine simulations in **Autodesk Fusion**.

## Projects

**Capstone: Early-Stage Wildfire Detection Network** McMaster University

*nRF52, BLE 5.0 Mesh, solar energy harvesting, low-power sensing* *Sept 2025 – Apr 2026*

- Designed a solar-powered **wireless sensor network** to catch wildfires at the smoldering stage: nodes with a $\sim$100 m range detect smoldering temperatures ($\sim$300°C) and relay heat level and location toward the lookout tower, while drones forward collected packets and imagery to a ground control station.
- Developed **nRF52-series firmware** for sensor sampling, local processing, short-term data buffering, and packet preparation and transmission over the integrated BLE radio.
- Implemented **BLE 5.0 Mesh** messaging over connectionless advertising packets (node ID, timestamp, sensor readings), avoiding power-hungry persistent connections while relaying data toward the tower under dense canopy.
- Designed a **solar energy-harvesting** power stage in which a boost charge controller stores low photodiode-array output in a **supercapacitor**, holding enough energy per charge for multiple packet transmissions.
- Integrated complementary low-power sensors: an **IR thermopile** for surface hotspots, an **electrochemical CO sensor** for smoldering-combustion gases, and a pulsed-LED **optical smoke sensor** for particulates.
- Co-designed the node's **housing structure** to ensure rugged operation in outdoor climates.